Q20 — LEXICAL ANALYZER

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


int main()

{

    char program[] =

        "#define PI 3.14\n"

        "#include<stdio.h>\n"

        "#include<conio.h>\n"

        "void main()\n"

        "{\n"

        "int a,b,c = 30;\n"

        "printf(\"hello\");\n"

        "}\n";


    int i = 0;

    int macros = 0;

    int headers = 0;

    printf("SOURCE PROGRAM:\n");

    printf("----------------\n");

    printf("%s", program);

    printf("\nCONSTANTS FOUND:\n");

    printf("----------------\n");

    while (program[i] != '\0')

    {

        /* Check for #define */
```

```
        if (program[i] == '#' &&
            strncmp(&program[i], "#define", 7) == 0)
        {
            macros++;
            i += 7;
        }

        /* Check for #include */
        else if (program[i] == '#' &&
                 strncmp(&program[i], "#include", 8) == 0)
        {
            headers++;
            i += 8;
        }

        /* Check for numeric constants */
        else if (isdigit(program[i]))
        {
            char number[20];
            int j = 0;

            while (isdigit(program[i]) || program[i] == '.')
            {
                number[j++] = program[i];
                i++;
    }

            number[j] = '\0';
```

```
            printf("%s\n", number);
        }
        else
        {
            i++;
        }
    }
    printf("\nNumber of Macros Defined = %d\n", macros);
    printf("Number of Header Files   = %d\n", headers);
    return 0;
}
```

OUTPUT

```
SOURCE PROGRAM:
----------------
#define PI 3.14
#include<stdio.h>
#include<conio.h>
void main()
{
int a,b,c = 30;
printf("hello");
}

CONSTANTS FOUND:
----------------
3.14
30

Number of Macros Defined = 1
Number of Header Files   = 2

--------------------------------
Process exited after 1.049 seconds with return value 0
Press any key to continue . . .
```